

2022-2023 高数学 A2 复习练习题 1

一、解答下列各题(每小题 5 分, 共 25 分)

1. 已知 $z = f(x, y)$, $x = \varphi(y, z)$, 其中 f, φ 均为可微函数, 求 $\frac{dz}{dx}$.
2. 求曲线 $x = \sin^2 t$, $y = \sin t \cos t$, $z = \cos^2 t$ 在对应于 $t = \frac{\pi}{4}$ 的点处的切线和法平面方程.
3. 设 $(\frac{ye^{-x}}{x}f(x) - \frac{2}{x^2})dx + (e^{-x}f(x) - \frac{4}{y^2})dy$ 是函数 $u(x, y)$ 的全微分, 其中 $f'(x)$ 连续且 $f(1) = e, u(1, 1) = 0$, 求 $u(x, y)$ 并计算积分: $\int_{(1,1)}^{(1,2)} (\frac{ye^{-x}}{x}f(x) - \frac{2}{x^2})dx + (e^{-x}f(x) - \frac{4}{y^2})dy$
4. 设 $a_n = \int_0^{n\pi} x|\sin x|dx$, 求级数 $\sum_{n=1}^{\infty} (\frac{1}{\sqrt{a_n}} - \frac{1}{\sqrt{a_{n+1}}})$ 的和.
5. 设 a, b 为非零向量, 且 $7a - 5b$ 与 $a + 3b$ 正交, 与 $a - 4b$ 与 $7a - 2b$ 正交, 求 $a^2 - b^2$.

二、解答下列各题(每小题 6 分, 共 30 分)

1. 设 $u = xy$, $v = \frac{x}{y}$, 试以新变量 u, v 变换方程 $x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = 0$, 其中 z 对各变量有二阶连续偏导数.
2. 设 $\vec{n}$ 是曲面 $z = x^2 + \frac{y^2}{2}$ 在点 $P(1, 2, 3)$ 处指向外侧的法向量, 求函数 $u = \sqrt{\frac{3x^2 + 3y^2 + z^2}{x}}$ 在 P 点处沿方向 $\vec{n}$ 的方向导数.
3. 改变二次积分 $\int_0^1 dx \int_0^x f(x, y)dy + \int_1^2 dx \int_0^{2-x} f(x, y)dy$ 的积分次序, 其中 $f(x, y)$ 连续.
4. 计算曲面积分 $\oiint_{\Sigma} (y + z - x)dydz + (x + y - z)dzdx + (x + y + z)dxdy$, 其中 Σ 是由 $|x| \leq 1, |y| \leq 1, |z| \leq 1$ 所确定的立体的表面外侧.
5. 设 $f(x, y) = \int_0^{xy} e^{-t^2} dt$, 求 $\frac{x}{y} \frac{\partial^2 f}{\partial x^2} - 2 \frac{\partial^2 f}{\partial x \partial y} + \frac{y}{x} \frac{\partial^2 f}{\partial y^2}$.

三、(5 分) 计算三重积分 $I = \iiint_{\Omega} z^2 dV$, 其中 Ω 由 $x^2 + y^2 + z^2 \leq 2z$ 所确定.

四、(5 分) 求半径为 R 的质量分布均匀的半球面的重心坐标.

五、(5 分) 已知曲面 S 的方程为 $S: x^2 + \frac{y^2}{4} + \frac{z^2}{2} = 1$, 平面 π 的方程为 $\pi: 2x + y + 2z + 6 = 0$.

(1) 求曲面 S 的平行于 π 的切平面方程; (2) 在曲面 S 上求到平面 π 距离为最短及最长的点, 并求最短及最长的距离.

六、(5 分) 设曲面方程为 $F(z - ax, z - by) = 0$ (a, b 为正常数). $F(u, v)$ 具有一阶连续的偏导

数, 且 $F_u^2 + F_v^2 \neq 0$, 试证明此曲面上任一点处法线恒垂直于一常向量.

七、(5 分) 试求通过直线 $L_1: \begin{cases} x-2z-4=0 \\ 3y-z+8=0 \end{cases}$ 且与直线 $L_2: x-1=y+1=z-3$ 平行的平面方程。

八、(5 分) 在第一卦限内作椭球面 $x^2 + y^2 + \frac{1}{4}z^2 = 1$ 的切平面, 使它在三个坐标轴上的截距平方和最小, 求该切平面的方程。

九、(5 分) 设 L 是光滑的正向简单闭曲线, 所围的区域记为 D , $\vec{n}$ 是 L 的单位外法线向量,

$u(x, y)$ 是具有二阶连续偏导数的二元函数, 试证:

$$\oint_L \frac{\partial u}{\partial n} ds = \iint_D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dxdy$$

十、(5 分) 已知 $u_n(x)$ 满足 $u_n'(x) = u_n(x) + x^{n-1}e^x$ (n 为正整数), 且 $u_n(1) = \frac{e}{n}$,

求函数项级数 $\sum_{n=1}^{\infty} u_n(x)$ 之和.

十一、(5 分) 设 $f(x)$ 是 $[-\pi, \pi]$ 上的偶函数, 且 $f(\frac{\pi}{2} + x) = -f(\frac{\pi}{2} - x)$, 证明: $f(x)$ 的余弦傅里叶级数展开式中 $a_{2n} = 0$.

参考答案

一、1. 利用全微分的不变形计算, 方程两边微分可得 $dz = f_x dx + f_y dy$, $dx = \varphi_y dy + \varphi_z dz$

$$\text{消去 } dy \text{ 可得 } \varphi_y dz - f_y dx = \varphi_y f_x dx - \varphi_z f_y dz \quad \frac{dz}{dx} = \frac{f_y + \varphi_y f_x}{\varphi_y + \varphi_z f_y};$$

2. 点 $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, 当 $t = \frac{\pi}{4}$ 时, $\vec{r} = (\dot{x}, \dot{y}, \dot{z}) = (1, 0, -1)$ 。

$$\text{切线 } \frac{x - \frac{1}{2}}{1} = \frac{y - \frac{1}{2}}{0} = \frac{z - \frac{1}{2}}{-1}, \text{ 法平面, } x - z = 0.$$

3. 解: 由于 $\left(\frac{ye^{-x}}{x} f(x) - \frac{2}{x^2}\right)dx + \left(e^{-x} f(x) - \frac{4}{y^2}\right)dy$ 是函数 $u(x, y)$ 的全微分, 所以 (x, y)

$$\text{不在坐标轴上时, 有 } \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}, \text{ 即 } \frac{\partial}{\partial x} \left(e^{-x} f(x) - \frac{4}{y^2}\right) = \frac{\partial}{\partial y} \left(\frac{ye^{-x}}{x} f(x) - \frac{2}{x^2}\right)$$

$$\text{从而得: } -e^{-x} f(x) + e^{-x} f'(x) = \frac{e^{-x}}{x} f(x), \text{ 即 } f'(x) = \left(\frac{1}{x} + 1\right)f(x)$$

$$\text{故有: } \frac{df(x)}{f(x)} = \left(\frac{1}{x} + 1\right)dx \Rightarrow \ln f(x) = \ln x + x + c$$

$$\text{而 } f(1) = e \Rightarrow c = 0, \text{ 因此 } \ln f(x) = \ln x + x \Rightarrow f(x) = e^{\ln x + x} = xe^x$$

$$\text{所以有 } \left(\frac{ye^{-x}}{x} f(x) - \frac{2}{x^2}\right)dx + \left(e^{-x} f(x) - \frac{4}{y^2}\right)dy = \left(y - \frac{2}{x^2}\right)dx + \left(x - \frac{4}{y^2}\right)dy$$

$$\text{故有: } \left(y - \frac{2}{x^2}\right)dx + \left(x - \frac{4}{y^2}\right)dy = ydx - \frac{2}{x^2}dx + xdy - \frac{4}{y^2}dy = d\left(xy + \frac{2}{x} + \frac{4}{y}\right)$$

$$\text{即 } u(x, y) = xy + \frac{2}{x} + \frac{4}{y} + c, \text{ 由 } u(1, 1) = 0, \text{ 得 } c = -7 \text{ 从而有 } u(x, y) = xy + \frac{2}{x} + \frac{4}{y} - 7$$

$$\text{故有: } \int_{(1,1)}^{(1,2)} \left(\frac{ye^{-x}}{x} f(x) - \frac{2}{x^2}\right)dx + \left(e^{-x} f(x) - \frac{4}{y^2}\right)dy = u(1, 2) - u(1, 1) = -1$$

$$\begin{aligned} \text{或 } \int_{(1,1)}^{(1,2)} \left(\frac{ye^{-x}}{x} f(x) - \frac{2}{x^2}\right)dx + \left(e^{-x} f(x) - \frac{4}{y^2}\right)dy \\ = \int_{(1,1)}^{(1,2)} \left(y - \frac{2}{x^2}\right)dx + \left(x - \frac{4}{y^2}\right)dy = \int_1^2 \left(1 - \frac{4}{y^2}\right)dy = \left(y + \frac{4}{y}\right) \Big|_1^2 = -1. \end{aligned}$$

$$4. \text{ 令 } x = n\pi - t, \text{ 则 } a_n = \int_0^{n\pi} (n\pi - t) |\sin t| dt = n\pi \int_0^{n\pi} |\sin t| dt - a_n$$

$$a_n = \frac{n\pi}{2} \int_0^{n\pi} |\sin t| dt = \frac{n^2 \pi}{2} \int_0^\pi |\sin t| dt = \frac{n^2 \pi}{2} \int_0^\pi \sin t dt = n^2 \pi \left[-\frac{1}{\sqrt{a_n}} + \frac{1}{\sqrt{a_{n+1}}} \right] = \frac{1}{\sqrt{\pi}} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$S_n = \sum_{k=1}^n \left(\frac{1}{\sqrt{a_k}} - \frac{1}{\sqrt{a_{k+1}}} \right) = \sum_{k=1}^n \frac{1}{\sqrt{\pi}} \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right) = \frac{1}{\sqrt{\pi}} \left(1 - \frac{1}{n+1} \right)$$

$$S = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{\pi}} \left(1 - \frac{1}{n+1} \right) = \frac{1}{\sqrt{\pi}}.$$

$$5. (7a - 5b) \cdot (a + 3b) = 0, (a - 4b) \cdot (7a - 2b) = 0.$$

$$\begin{cases} 7a^2 - 15b^2 + 16a \cdot b = 0 & (1) \\ 7a^2 + 8b^2 - 30a \cdot b = 0 & (2) \end{cases} \quad (1) \times 15 + (2) \times 8$$

$$161(a^2 - b^2) = 0, a^2 - b^2 = 0.$$

二、1. 解：不妨设 $z = f(u, v)$ ，则

$$dz = f_u du + f_v dv = f_u (ydx + xdy) + f_v \left(\frac{1}{y} dx - \frac{x}{y^2} dy \right) = \left(yf_u + \frac{f_v}{y} \right) dx + \left(xf_u - \frac{xf_v}{y^2} \right) dy$$

$$\frac{\partial z}{\partial x} = yf_u + \frac{f_v}{y}, \quad \frac{\partial z}{\partial y} = xf_u - \frac{xf_v}{y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = y \left(yf_{uu} + \frac{1}{y} f_{uv} \right) + \frac{1}{y} \left(yf_{uv} + \frac{1}{y} f_{vv} \right) = y^2 f_{uu} + 2f_{uv} + \frac{1}{y^2} f_{vv}$$

$$\frac{\partial^2 z}{\partial y^2} = x \left(xf_{uu} - \frac{x}{y^2} f_{uv} \right) + \frac{2x}{y^3} f_v - \frac{x}{y^2} \left(xf_{uv} - \frac{x}{y^2} f_{vv} \right) = x^2 f_{uu} - \frac{2x^2}{y^2} f_{uv} + \frac{x^2}{y^4} f_{vv} + \frac{2x}{y^3} f_v$$

$$x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = 4x^2 f_{uv} - \frac{2x}{y} f_v = 4uvf_{uv} - 2vf_v$$

在新变量 u, v 下方程 $x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = 0$ 可变为 $2uvf_{uv} - vf_v = 0$

2. 解： 设 $F(x, y, z) = x^2 + \frac{y^2}{2} - z$ $F_x = 2x, F_y = y, F_z = -1$

$F_x|_P = 2, F_y|_P = 2, F_z|_P = -1$ ，如图容易看出 $\vec{n}$ 与 z 正方向的夹角为钝角，其 z 轴坐标为

负，所以 $\vec{n} = \{2, 2, -1\}$ ， $\frac{\vec{n}}{|\vec{n}|} = \left\{ \frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right\}$

$$\frac{\partial u}{\partial x} = \frac{1}{2} \left(\frac{3x^2 + 3y^2 + z^2}{x} \right)^{-\frac{1}{2}} \frac{3x^2 - 3y^2 - z^2}{x^2}, \quad \frac{\partial u}{\partial y} = \frac{1}{2} \left(\frac{3x^2 + 3y^2 + z^2}{x} \right)^{-\frac{1}{2}} \frac{6y}{x}$$

$$\frac{\partial u}{\partial z} = \frac{1}{2} \left(\frac{3x^2 + 3y^2 + z^2}{x} \right)^{-\frac{1}{2}} \frac{2z}{x}$$

$$\frac{\partial u}{\partial x}|_P = -\frac{3}{4}\sqrt{6}, \quad \frac{\partial u}{\partial y}|_P = \frac{1}{2}\sqrt{6}, \quad \frac{\partial u}{\partial z}|_P = \frac{1}{4}\sqrt{6}$$

$$\frac{\partial u}{\partial l}|_P = -\frac{3}{4}\sqrt{6} \times \frac{2}{3} + \frac{\sqrt{6}}{2} \times \frac{2}{3} - \frac{\sqrt{6}}{4} \times \frac{1}{3} = -\frac{\sqrt{6}}{4}.$$

3. 原式 = $\int_0^1 dy \int_y^{2-y} f(x, y) dx$;

4. 由高斯公式得, 原式 = $\iiint_V dv = 8$.

5. $\frac{\partial f}{\partial x} = ye^{-x^2y^2}$, $\frac{\partial^2 f}{\partial x^2} = -2xy^3e^{-x^2y^2}$, $\frac{\partial^2 f}{\partial x \partial y} = e^{-x^2y^2} - 2x^2y^2e^{-x^2y^2}$

$\frac{\partial f}{\partial y} = xe^{-x^2y^2}$, $\frac{\partial^2 f}{\partial y^2} = -2x^3ye^{-x^2y^2}$,

$\frac{x}{y} \frac{\partial^2 f}{\partial x^2} - 2 \frac{\partial^2 f}{\partial x \partial y} + \frac{y}{x} \frac{\partial^2 f}{\partial y^2} = -2e^{-x^2y^2}$

三、 $I = \int_0^2 z^2 \pi(2z - z^2) dz = \frac{8}{5} \pi$

四、 设半球面方程为 $\Sigma: z = \sqrt{R^2 - x^2 - y^2}$, $\bar{z} = \frac{\iint_{\Sigma} z dS}{2\pi R^2} = \frac{R \iint_{D} dx dy}{2\pi R^2} = \frac{R}{2}$, 重心为 $(0, 0, \frac{R}{2})$.

五、 解 (1) S 上的点记为 (x, y, z) . S 的法向量 $(2x, \frac{y}{2}, z)$.

切平面与 π 平行, 则法向量对应坐标成比例: $\frac{2x}{2} = \frac{y/2}{1} = \frac{z}{2}$. $z = 2x, y = z$.

与曲面方程联立: $x^2 + x^2 + 2x^2 = 1, x = \pm \frac{1}{2}, y = \pm 1, z = \pm 1$.

切平面方程: $2x(X - x) + \frac{y}{2}(Y - y) + z(Z - z) = 0$,

利用曲面方程得 $2xX + \frac{y}{2}Y + zZ = 2, \pm X \pm \frac{1}{2}Y \pm Z = 2$.

平面 π 过点 $A = (-3, 0, 0)$, 点 $P_1 = (\frac{1}{2}, 1, 1), \overrightarrow{P_1A} = (-\frac{7}{2}, -1, -1)$,

P_1 到平面 π 的距离 $d_1 = \frac{|\overrightarrow{P_1A} \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|(-\frac{7}{2}, -1, -1) \cdot (2, 1, 2)|}{3} = \frac{10}{3}$.

点 $P_2 = (-\frac{1}{2}, -1, -1), \overrightarrow{P_2A} = (-\frac{5}{2}, 1, 1)$,

P_2 到平面 π 的距离 $d_2 = \frac{|\overrightarrow{P_2A} \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|(-\frac{5}{2}, 1, 1) \cdot (2, 1, 2)|}{3} = \frac{2}{3}$.

在曲面 S 上到平面 π 距离为最短及最长的点分别是 $(-\frac{1}{2}, -1, -1)$ 和 $(\frac{1}{2}, 1, 1)$,

并求最短及最长的距离分别是 $\frac{2}{3}$ 和 $\frac{10}{3}$.

六、 令 $z - ax = u$, $z - by = v$, 则, 曲面法向量 $\bar{n} = \{-aF_u, -bF_v, F_u + F_v\}$

取 $\bar{A} = \{b, a, ab\}$ 则 $\bar{n} \cdot \bar{A} = 0$, 从而 $\bar{n} \perp \bar{A}$.

七、解：由平面族方程知，所求平面方程有形式

$$\lambda_1(x-2z-4)+\lambda_2(3y-z+8)=0, \lambda_1x+3\lambda_2y+(-2\lambda_1-\lambda_2)z-4\lambda_1+8\lambda_2=0.$$

由于平面与 L_2 平行, $(\lambda_1, 3\lambda_2, -2\lambda_1-\lambda_2) \cdot (1, 1, 1) = 0, \lambda_1+3\lambda_2-2\lambda_1-\lambda_2=0, -\lambda_1+2\lambda_2=0.$

令 $\lambda_1=2, \lambda_2=1$, 得所求平面方程 $2x+3y-5z=0$.

八、解：设切点为 (a, b, c) ，法向量 $\vec{n} = (2a, 2b, \frac{c}{2})$ ，

$$\text{切平面方程为 } 2a(x-a)+2b(y-b)+\frac{c}{2}(z-c)=0,$$

$$\text{由 } a^2+b^2+\frac{c^2}{4}=1, \quad (1)$$

$$\text{得切平面方程为 } ax+by+\frac{c}{4}z=1,$$

$$\text{它在三个坐标轴上的截距平方和为 } f(a, b, c) = \frac{1}{a^2} + \frac{1}{b^2} + \frac{16}{c^2},$$

$$\text{作拉格朗日辅助函数 } L(a, b, c) = \frac{1}{a^2} + \frac{1}{b^2} + \frac{16}{c^2} + \lambda(a^2 + b^2 + \frac{c^2}{4} - 1),$$

$$\text{建立方程组 } \begin{cases} L_x = -\frac{2}{a^3} + 2\lambda a = 0 \\ L_y = -\frac{2}{b^3} + 2\lambda b = 0 \\ L_z = -\frac{32}{c^3} + \frac{\lambda c}{2} = 0 \\ L_\lambda = 0 \quad [\text{即(1)式}] \end{cases}$$

解得 $a=b=\frac{1}{2}, c=\sqrt{2}$ ，唯一可能极值点，此时截距平方和为最少，

$$\text{所求切平面方程为 } 2x+2y+\sqrt{2}z=4$$

$$\text{九、 } \frac{\partial u}{\partial n} = \left\{ \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right\} \cdot \vec{n}, \quad \frac{\partial u}{\partial n} ds = \left\{ \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right\} \cdot \{dy, -dx\} = -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$$

$$\oint_L \frac{\partial u}{\partial n} ds = \oint_L -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy = \iint_D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dxdy$$

十、解：先解一阶常系数微分方程，求出 $u_n(x)$ 的表达式，然后再求 $\sum_{n=1}^{\infty} u_n(x)$ 的和。

由已知条件可知 $u'_n(x) - u_n(x) = x^{n-1}e^x$ 是关于 $u_n(x)$ 的一个一阶常系数线性微分方程，故其通解为

$$u_n(x) = e^{\int dx} \left(\int x^{n-1} e^x e^{-\int dx} dx + c \right) = e^x \left(\frac{x^n}{n} + c \right),$$

$$\text{由条件 } u_n(1) = \frac{e}{n}, \text{ 得 } c = 0, \text{ 故 } u_n(x) = \frac{x^n e^x}{n},$$

$$\text{从而 } \sum_{n=1}^{\infty} u_n(x) = \sum_{n=1}^{\infty} \frac{x^n e^x}{n} = e^x \sum_{n=1}^{\infty} \frac{x^n}{n}.$$

$s(x) = \sum_{n=1}^{\infty} \frac{x^n}{n}$, 其收敛域为 $[-1, 1)$, 当 $x \in (-1, 1)$ 时, 有

$$s'(x) = \sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x},$$

$$\text{故 } s(x) = \int_0^x \frac{1}{1-t} dt = -\ln(1-x)$$

$$\text{当 } x = -1 \text{ 时, } \sum_{n=1}^{\infty} u_n(x) = -e^{-1} \ln 2.$$

于是, 当 $-1 \leq x < 1$ 时, 有 $\sum_{n=1}^{\infty} u_n(x) = -e^x \ln(1-x)$

十一、证 $a_{2n} = \frac{2}{\pi} \int_0^{\pi} f(x) \cos 2nxdx = \frac{2}{\pi} \left[\int_0^{\frac{\pi}{2}} f(x) \cos 2nxdx + \int_{\frac{\pi}{2}}^{\pi} f(x) \cos 2nxdx \right]$

$$\text{令 } t = \frac{\pi}{2} - x, \text{ 则}$$

$$\int_0^{\frac{\pi}{2}} f(x) \cos 2nxdx = -\int_{\frac{\pi}{2}}^0 f\left(\frac{\pi}{2} - t\right) \cos(n\pi - 2nt) dt = \int_0^{\frac{\pi}{2}} f\left(\frac{\pi}{2} - t\right) \cos(n\pi - 2nt) dt$$

$$\text{令 } t = x - \frac{\pi}{2}, \text{ 则}$$

$$\int_{\frac{\pi}{2}}^{\pi} f(x) \cos 2nxdx = \int_0^{\frac{\pi}{2}} f\left(\frac{\pi}{2} + t\right) \cos(n\pi + 2nt) dt$$

$$\text{因为 } f\left(\frac{\pi}{2} - t\right) = -f\left(\frac{\pi}{2} + t\right),$$

$$\cos(n\pi - 2nt) = \cos(-n\pi + 2nt) = \cos(-n\pi + 2nt + 2n\pi) = \cos(n\pi + 2nt)$$

$$\text{所以 } a_{2n} = \frac{2}{\pi} \left[-\int_0^{\frac{\pi}{2}} f\left(\frac{\pi}{2} + t\right) \cos(n\pi + 2nt) dt + \int_0^{\frac{\pi}{2}} f\left(\frac{\pi}{2} + t\right) \cos(n\pi + 2nt) dt \right] = 0$$

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